

Course 1 resources and citations

Module 1: Introduction to cloud computing

Resources

Helpful resources and tips

- [Resolve issues with a lab](#)
- [Google Docs Help Center](#)
- [Google Sheets Help Center](#)
- [How to use Google Slides](#)

Citations

Cloud computing infrastructure

- Attaran, M. (2017). [Cloud computing technology: Leveraging the power of the internet to improve business performance](#). *Journal of International Technology and Information Management*, 26(1).
- Conrad, J. (2022, May 23). [An introduction to cloud computing infrastructure](#). Capital One.
- Google Cloud. (n.d.). [Google Cloud infrastructure](#). Retrieved April 24, 2023.
- Google Cloud. (n.d.). [What is cloud computing?](#) Retrieved April 24, 2023.
- Sangle, A., & Achar, C. (2022). [Impact of cloud computing over traditional](#). *International Research Journal of Modernization in Engineering Technology and Science*, 04(06).

Cloud computing service models

- Attaran, M. (2017). [Cloud computing technology: Leveraging the power of the internet to improve business performance](#). *Journal of International Technology and Information Management*, 26(1).
- Conrad, J. (2022, May 23). [An introduction to cloud computing infrastructure](#). Capital One.
- Google Cloud. (n.d.). [Google Cloud infrastructure](#). Retrieved April 24, 2023.

- Google Cloud. (n.d.). [What is cloud computing?](#) Retrieved April 24, 2023.
- Sangle, A., & Achar, C. (2022). [Impact of cloud computing over traditional](#). *International Research Journal of Modernization in Engineering Technology and Science*, 04(06).

The history of cloud computing

- Foote, K. D. (2021, December 17). [A brief history of cloud computing](#). DATAVERSITY.

Benefits of cloud computing

- Google Cloud. (n.d.). [What is cloud computing?](#) Retrieved April 18, 2023.
- Google Cloud. (2023, November 15). [Google Cloud Platform services summary](#). Retrieved November 29, 2023.
- Sangle, A., & Achar, C. (2022). [Impact of cloud computing over traditional](#). *International Research Journal of Modernization in Engineering Technology and Science*, 04(06).

Traditional versus cloud computing

- Google Cloud. (n.d.). [What is cloud computing?](#) Retrieved April 18, 2023.
- Google Cloud. (2023, November 15). [Google Cloud Platform services summary](#). Retrieved November 29, 2023.
- Sangle, A., & Achar, C. (2022). [Impact of cloud computing over traditional](#). *International Research Journal of Modernization in Engineering Technology and Science*, 04(06).

Cloud data warehouses

- Google Cloud. (2023, November 29). [Overview: Migrate data warehouses to BigQuery](#).
- Google Cloud. (2024, January 18). [What is a data warehouse?](#)
- Maayan, G. D. (2018, April 30). [The difference between a traditional data warehouse and a cloud data warehouse](#). DATAVERSITY.

Introduction to BigQuery

- Google Cloud. (n.d.). [BigQuery](#). Retrieved May 2, 2023.
- Google Cloud. (2023, November 29). [Overview of BigQuery storage](#). Retrieved November 30, 2023.

- Thallam, R. (2020, September 2). [BigQuery explained: An overview of BigQuery's architecture](#). *Google Cloud Blog*.

Module 2: Data analytics in the cloud

Citations

Steps for effective cloud migration

- Google Cloud. (n.d.). [Application migration](#). Retrieved November 30, 2023.
- Google Cloud. (2023, October 9). [Migrate to Google Cloud: Get started](#).
- Google Cloud Tech. (2020, April 21). [Kickstarting your migration to Google Cloud \(5 steps\)](#) [Video]. YouTube.

Explore cloud deployment models

- Google Cloud. (n.d.). [What are the different types of cloud computing?](#) Retrieved November 29, 2023.
- Google Cloud. (n.d.). [What is a hybrid cloud?](#) Retrieved November 29, 2023.
- Google Cloud. (n.d.). [What is cloud computing?](#) Retrieved April 18, 2023.
- Google Cloud Tech. (2020, April 21). [Kickstarting your migration to Google Cloud \(5 steps\)](#) [Video]. YouTube.

Benefits of cloud deployment models

- Rasmussen, D. (n.d.). [An overview of cloud computing: Fundamentals, deployment models, and key concepts](#). Harvard University.

Cloud's role in advancing data analytics

- Google Cloud. (n.d.). [BigQuery](#). Retrieved May 2, 2023.
- Google Cloud. (n.d.). [What is a data warehouse?](#) Retrieved May 10, 2023.
- Google Cloud. (2023, November 29). [Overview: Migrate data warehouses to BigQuery](#). Retrieved November 30, 2023.

Benefits of cloud data analytics

- Google Cloud. (n.d.). [Advantages and disadvantages of cloud computing](#). Retrieved November 8, 2023.

Cost considerations of cloud services

- Google Cloud. (n.d.). [BigQuery pricing](#). Retrieved May 12, 2023.
- Google Cloud. (n.d.). [Cloud storage pricing](#). Retrieved May 12, 2023.
- Google Cloud. (n.d.). [Migrate to virtual machines: Cloud migration](#). Retrieved November 30, 2023.

Cloud data analytics in different industries

- Google Cloud. (n.d.). [Google Cloud for healthcare and life sciences](#). Retrieved November 30, 2023.
- Google Cloud. (n.d.). [Manufacturing Data Engine and Manufacturing Connect](#). Retrieved November 29, 2023.
- Kumar, M. (2022, January 19). [Impact of cloud computing in transportation industry](#). Aeologic.

Overview of cloud architecture

- Google Cloud. (n.d.). [What is cloud architecture?](#) Retrieved May 18, 2023.

The Google Cloud Architecture Framework

- Google Cloud. (2023, August 17). [Google cloud architecture framework](#). Retrieved October 25, 2023.

Cost optimization for the cloud

- Lerma, J., Sharma, P., Yadron, A., Sallaway, A., & Kumbhar, A. (n.d.). [Understanding the principles of cost optimization](#) [White paper]. Google Cloud.
- Sharma, P., & Nikl, T. (2023, September 28). [Five ways to optimize cloud spend with FinOps](#). Google Cloud Blog.

Cost optimization tips

- Google Cloud. (2023, November 29). [Cost optimization on Google Cloud for developers and operators](#). Retrieved November 30, 2023.

Module 3: The data life cycle

Resources

Importance of data management

- [Elements of a data management plan](#)
- [Writing a DMP step by step](#)

[Supplemental] Store and use data with SQL

- [SQL tutorial](#)

Summary of the data lifecycle

- [Data lifecycle management: Definition & 5 stages](#)
- [Introduction to the data lifecycle](#)

Overview of data team roles

- [How to structure your data analytics team](#)
- [Welcome to the data team handbook](#)

Types of automation for the data lifecycle

- [Options for controlling data lifecycles](#)
- [What is data analytics automation? \(Plus Benefits\)](#)

Versioning and holds for accurate data

- [Version control](#)
- [Version control - Research data management](#)

Citations

Introduction to data management

- Emerson College. (2021). [Data management tips for Google Workspace](#).
- Google Cloud. (n.d.). [What is data governance?](#) Retrieved November 8, 2023.
- The Turing Way Community. (2022). *Data management plan. The Turing Way: A handbook for reproducible, ethical and collaborative research*. Zenodo. doi: 10.5281/zenodo.3233853.

Importance of data management

- Google Cloud. (2023, August 8). [Analyze your data](#). Retrieved November 28, 2023.
- Lazzari, T. de, Akhtar, S. N., Montag, D., Ribbrock, A., Wells, Z., & Tekiner, F. (2023). [Three pillars for building a modern data strategy](#). Google Cloud.

Safety and privacy in the cloud

- Google Cloud. (n.d.). [Compliance resource center](#). Retrieved April 28, 2023.
- Google Cloud. (n.d.). [Google Cloud & the general data protection regulation \(GDPR\)](#). Retrieved May 3, 2023.
- Google Cloud. (n.d.). [Security and identity](#). Retrieved April 28, 2023.
- Google Cloud. (n.d.). [What is encryption?](#) Retrieved May 3, 2023.
- Google Data Centers. (n.d.). [Data and security](#). Retrieved April 28, 2023.
- Google Help Center. (n.d.). [Understanding PII in Google's contracts and policies](#). Analytics Help. Retrieved April 28, 2023.

Stages of the data lifecycle

- Yi, M. (n.d.). [Distinguishing data roles: Engineers, analysts, and scientists](#). Chartio.

Summary of the data lifecycle

- Google Cloud. (2023, August 8). [Analyze your data](#). Retrieved November 28, 2023.

The data lifecycle in action

- Google Cloud. (2023, August 8). [Analyze your data](#). Retrieved November 28, 2023.

Common roles on a data team

- Berkeley Extension. (2019, October 18). [Understanding Data Science Roles: Who Does What?](#) Berkeley Boot Camps.
- Brown, S. (2020, May 27). [How to build a data analytics dream team](#). MIT Management Sloan School.
- Stobierski, T. (2021, March 9). [How to structure your data analytics team](#). Harvard Business School Online.

Overview of data team roles

- Berkeley Extension. (2019, October 18). [Understanding Data Science Roles: Who Does What?](#) *Berkeley Boot Camps*.
- Stobierski, T. (2021, March 9). [How to structure your data analytics team](#). *Harvard Business School Online*.

[Supplemental] Overview of data analysis and business intelligence

- Merriam-Webster. (n.d.). [Merriam-Webster.com dictionary](#). Retrieved November 7, 2023.
- University of Houston Library. (2021, September 1). [What is Data?](#) [Video]. YouTube.

Importance of automation in the data lifecycle

- George, J., & Kraaz, M. (2023, September 22). [How “Anything is Possible” automated data pipelines with BigQuery and Windsor.ai](#). *Google Cloud Blog*.
- Google Cloud. (2023, July 17). [Create a culture of automation](#). Retrieved November 28, 2023.
- Schlesinger, S., & Gavzy, A. (2021, July 12). [Automation of the data lifecycle: Focus on data creation](#). RTInsights.

Types of automation for the data lifecycle

- Coursera. (2023, October 26). [3 types of machine learning you should know](#).
- Jeremiah, O. (2023, March 16). [How to generate automated reports from a SQL database using Python](#). freeCodeCamp.

Introduction to data retention policies

- Barney, N. (2023, June). [PCI DSS \(Payment Card Industry Data Security Standard\)](#). Tech Target.
- Google Cloud. (n.d.). [Bucket lock](#).
- Office for Civil Rights (OCR). (2022, March 31). [The HIPAA Privacy Rule](#). US Department of Health and Human Services.
- Wagner, S., & Dittmar, L. (2006, April 1). [The unexpected benefits of sarbanes-oxley](#). Harvard Business Review.

Overview of version control and holds

- Google Cloud Tech. (2019, December 15). [Dealing with and managing versioning](#) [Video]. YouTube.
- Google Vault Help. (n.d.). [Get started with holds in Google Vault](#). Retrieved May 21, 2023.

Versioning and holds for accurate data

- Google Cloud. (2023, November 29). [Version control your code](#). Retrieved November 29, 2023.
- Google Vault Help. (n.d.). [Get started with holds in Google Vault](#). Retrieved September 22, 2023.

Module 4: The role of a cloud data analyst

Resources

Data analyst job responsibilities

- [A day in the life of a data analyst—What's It really like?](#)
- [What does a data analyst do? Exploring the day-to-day of this tech career](#)

End-to-end data management

- [10 best practices for analytics success \(including 3 you can't ignore\)](#)
- [Reimagine data analytics for the era of AI](#)

Types of data access management

- [IAM: There and back again using resource hierarchies](#)
- [Tips on building a network security policy in Google Cloud](#)

Communicate and collaborate with stakeholders

- [How to create a stakeholder engagement plan](#)
- [Identifying and analyzing stakeholders and their interests](#)

The data request process

- [Managing data requests](#)

- [Responding to data requests](#)

Google Cloud's data storage and management tools

- [Go from prototype to production quickly](#)

Uses of generative AI in data analytics

- [Get started with generic search](#)
- [How AI is rewriting the rules of data analysis](#)

Guide to Google Cloud

- [Cloud Billing documentation](#)
- [Free cloud features and trial offer](#)
- [Google Cloud console](#)
- [Google Cloud console overview](#)
- [Why Google Cloud](#)

Citations

Data analysis in the cloud

- Google Cloud. (n.d.). [Drive innovation with Google Cloud smart analytics solutions](#). Retrieved November 28, 2023.
- Google Cloud. (2023, November 27). [Overview of BigQuery analytics](#). Retrieved November 28, 2023.
- Google Cloud Tech. (2023, February 3). [Google Cloud data analytics in 10 minutes](#) [Video]. YouTube.

Data analyst job responsibilities

- Google Cloud. (n.d.). [What is cloud analytics?](#) Retrieved November 29, 2023.
- Joubert, S. (2019, April 17). [What does a data analyst do?](#) Northeastern University.

Overview of Google's cloud data analytics tools

- Google Cloud. (n.d.). [Cloud data fusion](#). Retrieved May 30, 2023.

- Google Cloud. (n.d.). [BigQuery](#). Retrieved May 30, 2023.
- Google Cloud. (n.d.). [Dataflow](#). Retrieved May 30, 2023.
- Google Cloud. (n.d.). [Knowledge Catalog](#). Retrieved May 30, 2023.
- Google Cloud. (n.d.). [Managed Service for Apache Spark](#). Retrieved May 30, 2023.
- Google Cloud. (n.d.). [Google Cloud products](#). Retrieved May 26, 2023.
- Google Cloud. (n.d.). [Looker](#). Retrieved May 26, 2023.

End-to-end data management

- Google Cloud. (2021, December 10). [Google Cloud's end-to-end data cloud demo](#) [Video]. YouTube.
- Google Cloud. (2023, November 29). [Work with Dataflow data pipelines](#). Retrieved November 30, 2023.

Data access management for secure data

- Google Cloud. (2024, January 17) [IAM overview](#).
- Google Cloud. (n.d.). [Identity and Access Management \(IAM\)](#). Retrieved May 30, 2023.
- Google Cloud. (2023, November 27). [Identity and access management \(IAM\)](#). Retrieved November 28, 2023.

Types of data access management

- Google Cloud. (2023, November 29). [Overview of access control](#). Retrieved November 30, 2023.
- Google Cloud. (2023, November 27). [Identity and access management \(IAM\)](#). Retrieved November 30, 2023.

Introduction to business data requests

- Education Commission of the States & The National Endowment for the Arts. (2020). [The Arts Education Data Toolkit](#) (pp. 32-38). Education Commission of the States.
- Skylight. (n.d.). [Preparing a successful data request](#).
- Statewide Longitudinal Data Systems (SLDS) Grant Program & National Center for Education Statistics. (2019, May 29). [SLDS Topical Webinar Summary: Managing data requests](#) (PDF). Institute of Education Sciences.

Communicate and collaborate with stakeholders

- LinkedIn. (2023, August 17). [How do you coordinate with data analysts and stakeholders online?](#)
- Statewide Longitudinal Data Systems (SLDS) Grant Program & National Center for Education Statistics. (2019, May 29). [SLDS Topical Webinar Summary: Managing data requests](#) (PDF). Institute of Education Sciences.

Tips for ensuring accurate data

- Harvard Business School. (2017, April 5). [How to analyze a dataset: 6 steps](#). *Business Insights Blog*.
- LinkedIn. (2023, October 13). [How do you integrate data from multiple sources into your data analysis?](#)
- Longwood Research Data Management. (n.d.). [Analysis ready datasets](#). Harvard Medical School.
- University of Texas at Austin. (n.d.). [Outliers](#). Statistics Online Support.

The data request process

- LinkedIn. (2023, August 3). [How do you manage data requests from partners?](#)
- Skylight. (n.d.). [Responding to data requests](#).

Cloud data storage and management tools

- Dickinson, E. (2020, November 25). [Wayfair delights suppliers and customers with help from Google Cloud](#). *Google Cloud Blog*.
- Google Cloud. (2024, January 18). [Cloud data fusion overview](#).
- Google Cloud. (n.d.). [Dataflow](#). Retrieved June 4, 2023.
- Google Cloud. (n.d.). [What is cloud storage?](#) Retrieved June 4, 2023.
- Howe-Ely, M. (2020, September 24). [How to combine data from two BigQuery public datasets using SQL](#). Michael Howe-Ely.

Google Cloud's data storage and management tools

- Google Cloud. (n.d.). [Cloud Data Fusion](#). Retrieved September 22, 2023.
- Google Cloud. (n.d.). [BigQuery](#). Retrieved September 22, 2023.

- Google Cloud. (n.d.). [Dataflow](#). Retrieved September 22, 2023.
- Google Cloud. (n.d.). [Managed Service for Apache Spark](#). Retrieved September 23, 2023.
- Google Cloud. (n.d.). [Cloud Storage](#). Retrieved September 22, 2023.

Introduction to Managed Service for Apache Spark

- Dickinson, E. (2020, November 25). [Wayfair delights suppliers and customers with help from Google Cloud](#). *Google Cloud Blog*.
- Google Cloud. (n.d.). [Managed Service for Apache Spark](#). Retrieved June 30, 2023.
- Informatica. (n.d.). [Ephemeral clusters](#).

Uses of generative AI in data analytics

- Google Cloud. (n.d.). [Agent Search on Gemini Enterprise Agent Platform](#). Retrieved November 30, 2023.
- Gupta, A. (2020, September 29). [Creating intelligent bots with data analytics capabilities and beyond—A working model](#). LinkedIn.
- Stafford, G. (2023, May 31). [Generative AI for analytics: Performing natural language queries on Amazon RDS using Sagemaker, Langchain, and LLMs](#). LinkedIn.

Strategies for handling data requests

- Google Issue Tracker. (2022, May 22). [Parent-child relationships](#). Retrieved June 19, 2023.
- Google Issue Tracker. (2023, October 19). [Issues](#). Retrieved June 19, 2023.
- Statewide Longitudinal Data Systems (SLDS) Grant Program & National Center for Education Statistics. (2019, May 29). [SLDS Topical Webinar Summary: Managing data requests](#) (PDF). Institute of Education Sciences.

Elements of a data request

- Education Commission of the States & The National Endowment for the Arts. (2020). [The Arts Education Data Toolkit](#) (pp. 32-38). Education Commission of the States.
- Statewide Longitudinal Data Systems (SLDS) Grant Program & National Center for Education Statistics. (2019, May 29). [SLDS Topical Webinar Summary: Managing data requests](#) (PDF). Institute of Education Sciences.

Clear documentation benefits the whole team

- Barrett, D. (2022, August 12). [A data playbook](#). *Medium*.
- Blisle, P. (2015, May 13). [Codebook cookbook: How to enter and document your data](#). McGill.
- California Health and Human Services Agency. (n.d.) [Five plays for your Department](#). California Health and Human Services Agency Data Playbook 3rded.
- Illinois Library. (2022, September 6). [Introduction to data management for undergraduate students: Data documentation](#).
- Social Science Research Council & The Qualitative Data Repository. (n.d.). [*Interactive online course notes on [Principles of Documenting Data](#)*]. Managing Qualitative Social Science Data, Social Science Research Council.
- University of Arizona. (n.d.). [Data documentation](#).